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Indian Institute of Technology Hyderabad

Stochastic Processes

Lecture 11

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IID Process

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let $\mathbf{X} = \{X_t : t \in \mathbb{R}_+\}$ be a random process defined w.r.t. $\mathscr{F}$.

Definition (IID Process)

The process $\mathbf{X} = \{X_t : t \in \mathbb{R}_+\}$ is said to be an **IID process** with the **common CDF F** if for any $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathbb{R}_+^n$, and $\mathbf{x} = (x_1, \ldots, x_n)$,

$$F_{\mathbf{t}}(\mathbf{x}) = F_{X_{t_1}, \ldots, X_{t_n}}(x_1, \ldots, x_n) = \prod_{i=1}^{n} F_{X_{t_i}}(x_i) = \prod_{i=1}^{n} F(x_i).$$

Exercises:

- The FDDs of $\mathbf{X} = \{X_t : t \in \mathbb{R}_+\}$ are consistent.
- $\mathbf{X} = \{X_t : t \in \mathbb{R}_+\}$ is strictly stationary. That is, **every IID process is stationary**.

Example: Random Walk

Let $\{X_n\}_{n\in\mathbb{N}}$ be an $\mathbb{R}$-valued **IID process** with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 < +\infty$.
Let $S_0 := 0$, and for each $t \in \mathbb{N}$, let

$$S_t = \sum_{i=1}^{t} X_i.$$

The index set for the random walk process $\{S_t\}_{t\geq 0}$ is $\mathcal{T} = \mathbb{N} \cup \{0\}$

- Determine M_S and C_S for the process $\{S_t\}_{t\in\mathcal{T}}$.
- Is $\{S_t\}_{t\in\mathcal{T}}$ wide-sense stationary?
- If $X_1, X_2, \ldots$ are discrete random variables, determine the joint PMF of $S_1, \ldots, S_t$ for a fixed $t \geq 1$.
- If $X_1, X_2, \ldots$ are continuous random variables, determine the joint PDF of $S_1, \ldots, S_t$ for a fixed $t \geq 1$.
- Show that $\{S_t\}_{t\in\mathcal{T}}$ has **stationary increments**, i.e., for any $n \in \mathbb{N}$ and $0 = t_0 \leq t_1 < t_2 < \cdots < t_n$,

$$(S_{t_1} - S_{t_0},\ S_{t_2} - S_{t_1},\ \ldots,\ S_{t_n} - S_{t_{n-1}}) \overset{\text{d.}}{=} (S_{t_1+h} - S_{t_0+h},\ S_{t_2+h} - S_{t_1+h},\ \ldots,\ S_{t_n+h} - S_{t_{n-1}+h}) \qquad \forall h \in \mathcal{T}.$$

- Show that $\{S_t\}_{t\in\mathcal{T}}$ has **independent increments**, i.e., for any $n \in \mathbb{N}$ and $0 = t_0 \leq t_1 < t_2 < \cdots < t_n$,

$$S_{t_1} - S_{t_0},\ S_{t_2} - S_{t_1},\ \ldots,\ S_{t_n} - S_{t_{n-1}} \quad \text{are independent.}$$

Example: Gaussian Process

- Let $\{X_t : t \in \mathbb{R}\}$ be a continuous-time process.
  For any $n \in \mathbb{N}$ and $\mathbf{t} = (t_1, \ldots, t_n) \in \mathbb{R}^n$, let $(X_{t_1}, \ldots, X_{t_n})$ be **jointly Gaussian**:
  for any $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$,

$$f_{X_{t_1},\ldots,X_{t_n}}(x_1, \ldots, x_n) \;=\; f_{\mathbf{t}}(\mathbf{x}) \;=\; \frac{1}{\sqrt{(2\pi)^n \det(K_{\mathbf{t}})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{\mathbf{t}})^\top K_{\mathbf{t}}^{-1}(\mathbf{x} - \boldsymbol{\mu}_{\mathbf{t}})\right),$$

  where $\boldsymbol{\mu}_{\mathbf{t}}$ and $K_{\mathbf{t}}$ are given by

$$\boldsymbol{\mu}_{\mathbf{t}} \;=\; \begin{bmatrix} \mathbb{E}[X_{t_1}] \\ \vdots \\ \mathbb{E}[X_{t_n}] \end{bmatrix} \;=\; \begin{bmatrix} M_X(t_1) \\ \vdots \\ M_X(t_n) \end{bmatrix}, \qquad K_{\mathbf{t}} \;=\; \begin{bmatrix} C_X(t_1,t_1) & C_X(t_1,t_2) & \cdots & C_X(t_1,t_n) \\ C_X(t_2,t_1) & C_X(t_2,t_2) & \cdots & C_X(t_2,t_n) \\ \vdots & \vdots & \vdots & \vdots \\ C_X(t_n,t_1) & C_X(t_n,t_2) & \cdots & C_X(t_n,t_n) \end{bmatrix}$$

Exercise:

A wide-sense stationary Gaussian process is stationary, i.e., for any $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathbb{R}^n$, $h \in \mathbb{R}$,

$$f_{\mathbf{t}+h} \;=\; f_{\mathbf{t}}.$$

Filtration and Stopping Times

σ-Algebra Generated by a Random Variable

Fix a measurable space $(\Omega, \mathscr{F})$.
Let $X : \Omega \to \mathbb{R}$ be a random variable w.r.t. $\mathscr{F}$.

Definition (σ-Algebra Generated by a Random Variable)

The **σ-algebra generated by X**, denoted $\sigma(X)$, is defined as

$$\sigma(X) := \left\{ A \in \mathscr{F} : \ A = X^{-1}(B) \text{ for some } B \in \mathscr{B}(\mathbb{R}) \right\}.$$

- $\sigma(X)$ is the collection of all events whose occurrence can be determined by observing X
- $\sigma(X)$ is the smallest σ-algebra w.r.t. which X will be measurable

σ-Algebra Generated by a Random Vector

Fix a measurable space $(\Omega, \mathscr{F})$.
Let $(X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n$ be a random vector w.r.t. $\mathscr{F}$.

Definition (σ-Algebra Generated by a Random Vector)

The **σ-algebra generated by $(X_1, \ldots, X_n)$**, denoted $\sigma(X_1, \ldots, X_n)$, is defined as

$$\sigma(X_1, \ldots, X_n) := \left\{ A \in \mathscr{F} : A = (X_1, \ldots, X_n)^{-1}(B) \text{ for some } B \in \mathscr{B}(\mathbb{R}^n) \right\}.$$

- For each $n \in \mathbb{N}$, let
$$\mathscr{G}_n \;:=\; \sigma(X_1, \ldots, X_n).$$
- Then, observe that
$$\mathscr{G}_1 \subseteq \mathscr{G}_2 \subseteq \mathscr{G}_3 \subseteq \cdots, \qquad \mathscr{G}_n \subseteq \mathscr{F} \quad \forall\, n.$$

Filtration

Fix a measurable space $(\Omega, \mathscr{F})$. Let $\mathcal{T}$ be an **ordered**[1] index set.

Definition (Filtration)

A collection of σ-algebras $\mathscr{G} = \{\mathscr{G}_t : t \in \mathcal{T}\}$ with $\mathscr{G}_t \subseteq \mathscr{F}$ for all $t \in \mathcal{T}$ is called a **filtration** if

$$\mathscr{G}_s \subseteq \mathscr{G}_t \qquad \forall\, s \leq t.$$

Examples of filtration:

- Let $\mathbf{X} = \{X_t : t \in \mathcal{T}\}$ be a stochastic process defined w.r.t. $\mathscr{F}$. Let

$$\mathscr{G}_t^{\mathbf{X}} := \sigma\Big(\{X_s : s \leq t\}\Big), \qquad t \in \mathcal{T}.$$

  Then, $\mathscr{G}^{\mathbf{X}} = \{\mathscr{G}_t^{\mathbf{X}} : t \in \mathcal{T}\}$ is a filtration, called the **natural filtration** adapted to the process $\mathbf{X}$
- $\mathscr{G}_t^{\mathbf{X}}$ represents the "information flow" in the process $\mathbf{X}$ up to time t

[1] Partial ordering suffices.

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Examples of filtration:

- Let $\mathbf{X} = \{X_t : t \in \mathcal{T}\}$ and Y be defined w.r.t. $\mathscr{F}$. Let $\mathscr{G}_t^{\mathbf{X}} = \sigma\Big(\{X_s : s \leq t\}\Big)$, and let

$$\mathscr{F}_t := \mathscr{G}_t^{\mathbf{X}} \vee \sigma(Y) = \sigma\Big(\mathscr{G}_t^{\mathbf{X}} \cup \sigma(Y)\Big), \qquad t \in \mathcal{T}.$$

  Then, $\{\mathscr{F}_t : t \in \mathcal{T}\}$ is a filtration
- Y typically represents "side information" or "insider information"
- $\mathscr{F}_t$ represents the information available at time t to an agent
  - who is observing the process $\mathbf{X}$, and
  - has insider knowledge Y

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$$\mathscr{F}_t := \sigma\Big(\mathscr{G}_t^{\mathbf{X}} \vee \sigma(\tau \wedge t)\Big), \qquad t \in \mathcal{T}.$$

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- You can think of τ as a random time instant at which system behaviour changes
- $\mathscr{F}_t$ encodes the information from $\mathbf{X}$ seen up to time t, while also keeping track of whether behaviour change has occurred as of t
  - If change has occurred before t, you know exactly when change has occurred

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Stochastic Processes

Lecture 12

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20 February 2026

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Stopping Time

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathcal{T} = \mathbb{R}_+$. Let $\mathscr{G} = \{\mathscr{G}_t : t \in \mathcal{T}\}$ be a filtration.

Definition (Stopping Time)

A random variable $\tau : \Omega \to \mathbb{R}_+ \cup \{+\infty\}$ is said to be a **stopping time adapted to the filtration $\mathscr{G}$** if:

- $\mathbb{P}\Big(\Big\{\tau < +\infty\Big\}\Big) = 1.$
- For each $t \in \mathcal{T}$,

$$\{\tau \leq t\} \in \mathscr{G}_t.$$

Interpretation of stopping time:

- If τ is a stopping time adapted to $\{\mathscr{G}_t : t \in \mathcal{T}\}$, then for any $t \in \mathcal{T}$, the answer to the question

$$\text{“ Is } \tau \leq t \text{ ? ”}$$

can be determined based only on the information in $\mathscr{G}_t$

Example

- Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be an IID, real-valued process.
  Fix $A \subseteq \mathscr{B}(\mathbb{R})$ with $\mathbb{P}(\{X_1 \in A\}) = p_A \in (0, 1)$.
  Let $\tau_{\mathbf{X}}^A$ be defined as
  $$\tau_{\mathbf{X}}^A \;:=\; \inf\{n \geq 1 : X_n \in A\}.$$
  Show that $\tau_{\mathbf{X}}^A$ is a stopping time adapted to the natural filtration of $\mathbf{X}$.

Discrete Stopping Times

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n : n \in \mathbb{N}\}$ be a process w.r.t. $\mathscr{F}$.
Consider the natural filtration $\mathscr{G}^{\mathbf{X}} = \{\mathscr{G}_n^{\mathbf{X}} : n \in \mathbb{N}\}$, where $\mathscr{G}_n^{\mathbf{X}} = \sigma(X_1, \ldots, X_n)$ for each n.
Let $\tau : \Omega \to \mathbb{N} \cup \{+\infty\}$ be a RV w.r.t. $\mathscr{F}$.

Lemma (Discrete Stopping Times)

The RV τ is a stopping time adapted to $\mathscr{G}^{\mathbf{X}}$ **if and only if**

$$\mathbb{P}(\{\tau < +\infty\}) = 1, \qquad \{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}} \quad \forall\, n \in \mathbb{N}.$$

Proof of Lemma 1:

- **(If part):** Suppose $\mathbb{P}(\{\tau < +\infty\}) = 1$ and $\{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}}$ for each n
  - We need to show that $\{\tau \leq n\} \in \mathscr{G}_n^{\mathbf{X}}$ for each n
  - For any $n \in \mathbb{N}$, we have

$$\{\tau \leq n\} = \bigcup_{i=1}^{n} \{\tau = i\}.$$

  - $\{\tau = 1\} \in \mathscr{G}_1^{\mathbf{X}} \subseteq \mathscr{G}_n^{\mathbf{X}} \quad \ldots \quad \{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}}$; therefore, $\bigcup_{i=1}^{n} \{\tau = i\} \in \mathscr{G}_n$

Discrete Stopping Times

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n : n \in \mathbb{N}\}$ be a process w.r.t. $\mathscr{F}$.
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Proof of Lemma 1:

- **(Only if part):** Suppose $\mathbb{P}(\{\tau < +\infty\}) = 1$ and $\{\tau \leq n\} \in \mathscr{G}_n^{\mathbf{X}}$ for each n
  - We need to show that $\{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}}$ for each n
  - For any $n \in \mathbb{N}$, we have
  $$\{\tau = n\} \;=\; \{\tau \leq n\} \setminus \{\tau \leq n-1\}.$$
  - $\{\tau \leq n-1\} \in \mathscr{G}_{n-1}^{\mathbf{X}} \subseteq \mathscr{G}_n^{\mathbf{X}}$, $\quad \{\tau \leq n\} \in \mathscr{G}_n^{\mathbf{X}}$; $\quad$ therefore, $\{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}}$

Wald's Identity (Wald's Lemma)

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Lemma (Wald's Identity)

Let $\mathbf{X} = \{X_t\}_{t \in \mathbb{N}}$ *be an* **IID process** *with* $\mathbb{E}\left[\left|X_1\right|\right] < +\infty$. *For each* $t \in \mathbb{N}$, *let*

$$S_t = \sum_{i=1}^{t} X_i.$$

If τ *is a* **stopping time** *adapted to the natural filtration of the process* $\mathbf{X}$ *with* $\mathbb{E}[\tau] < +\infty$, *then*

$$\mathbb{E}[S_\tau] \;=\; \mathbb{E}\left[\sum_{i=1}^{\tau} X_i\right] \;=\; \mathbb{E}[\tau] \cdot \mathbb{E}[X_1].$$

Proof of Lemma 3

$$
\begin{aligned}
\mathbb{E}[S_\tau] &= \mathbb{E}\left[\sum_{i=1}^{\tau} X_i\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} X_i \, \mathbf{1}_{\{\tau \geq i\}}\right] \\
&\overset{(a)}{=} \sum_{i=1}^{\infty} \mathbb{E}\left[X_i \, \mathbf{1}_{\{\tau \geq i\}}\right] \\
&= \sum_{i=1}^{\infty} \mathbb{E}[X_i] \cdot \mathbb{E}\left[\mathbf{1}_{\{\tau \geq i\}}\right] \\
&= \mathbb{E}[X_1] \cdot \sum_{i=1}^{\infty} \mathbb{E}\left[\mathbf{1}_{\{\tau \geq i\}}\right] \overset{(b)}{=} \mathbb{E}[X_1] \cdot \mathbb{E}\left[\sum_{i=1}^{\infty} \mathbf{1}_{\{\tau \geq i\}}\right] \overset{(c)}{=} \mathbb{E}[X_1] \cdot \mathbb{E}[\tau].
\end{aligned}
$$

(a), (b) – follow from MCT & $\mathbb{E}\left[\left|X_1\right|\right] < +\infty$, $\quad$ (c) – follows from $\mathbb{E}[\tau] < +\infty$

Example

- Suppose $X_1, X_2, \cdots \overset{\text{i.i.d.}}{\sim} \text{Geometric}(0.5)$.
  For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.
  Let τ be defined as
  $$\tau := \inf\left\{n \geq 1 : S_n = 33\right\}.$$
  Determine $\mathbb{E}[\tau]$.

Example

- Suppose $X_1, X_2, \cdots \overset{\text{i.i.d.}}{\sim} \mathcal{N}(1,1)$.
  For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.
  Let τ be defined as
  $$\tau := \inf\left\{n \geq 1 : S_n = \frac{\pi}{2}\right\}.$$
  Determine $\mathbb{E}[\tau]$.

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Lecture 13

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24 February 2026

Stopping Time

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathcal{T} = \mathbb{R}_+$. Let $\mathscr{G} = \{\mathscr{G}_t : t \in \mathcal{T}\}$ be a filtration.

Definition (Stopping Time)

A random variable $\tau : \Omega \to \mathbb{R}_+ \cup \{+\infty\}$ is said to be a **stopping time adapted to the filtration $\mathscr{G}$** if:

- $\mathbb{P}\Big(\Big\{\tau < +\infty\Big\}\Big) = 1.$
- For each $t \in \mathcal{T}$,

$$\{\tau \leq t\} \in \mathscr{G}_t.$$

Interpretation of stopping time:

- If τ is a stopping time adapted to $\{\mathscr{G}_t : t \in \mathcal{T}\}$, then for any $t \in \mathcal{T}$, the answer to the question

$$\text{“ Is } \tau \leq t \text{ ? ”}$$

can be determined based only on the information in $\mathscr{G}_t$

Discrete Stopping Times

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n : n \in \mathbb{N}\}$ be a process w.r.t. $\mathscr{F}$.
Consider the natural filtration $\mathscr{G}^{\mathbf{X}} = \{\mathscr{G}_n^{\mathbf{X}} : n \in \mathbb{N}\}$, where $\mathscr{G}_n^{\mathbf{X}} = \sigma(X_1, \ldots, X_n)$ for each n.
Let $\tau : \Omega \to \mathbb{N} \cup \{+\infty\}$ be a RV w.r.t. $\mathscr{F}$.

Lemma (Discrete Stopping Times)

The RV τ is a stopping time adapted to $\mathscr{G}^{\mathbf{X}}$ **if and only if**

$$\mathbb{P}(\{\tau < +\infty\}) = 1, \qquad \{\tau = n\} \in \mathscr{G}_n^{\mathbf{X}} \quad \forall\, n \in \mathbb{N}.$$

Wald's Identity (Wald's Lemma)

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Lemma (Wald's Identity)

Let $\mathbf{X} = \{X_t\}_{t \in \mathbb{N}}$ *be an* **IID process** *with* $\mathbb{E}\Big[\big|X_1\big|\Big] < +\infty$. *For each* $t \in \mathbb{N}$, *let*

$$S_t = \sum_{i=1}^{t} X_i.$$

If $\tau : \Omega \to \mathbb{N} \cup \{+\infty\}$ *is a* **stopping time** *adapted to the natural filtration of the process* $\mathbf{X}$ *with* $\mathbb{E}[\tau] < +\infty$, *then*

$$\mathbb{E}[S_\tau] \;=\; \mathbb{E}\left[\sum_{i=1}^{\tau} X_i\right] \;=\; \mathbb{E}[\tau] \cdot \mathbb{E}[X_1].$$

Example: Wald's Lemma Not Applicable

- Suppose $X_1, X_2, \cdots \overset{\text{i.i.d.}}{\sim} \text{Geometric}(0.5)$.
  For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.
  Let τ be defined as
  $$\tau := \inf\left\{n \geq 1 : S_n = 33\right\}.$$
  Determine $\mathbb{E}[\tau]$.
- Observe that
  $$\{S_1 > 33\} \;\subseteq\; \{\tau = +\infty\}.$$
  Therefore, we have
  $$\mathbb{P}(\{\tau = +\infty\}) \;\geq\; \mathbb{P}(\{S_1 > 33\}) \;=\; (1/2)^{33} \;>\; 0.$$
  Thus, it follows that
  $$\mathbb{E}[\tau] \;=\; \sum_{n \in \mathbb{N}} n\,\mathbb{P}(\{\tau = n\}) \;+\; (+\infty)\,\mathbb{P}(\{\tau = +\infty\}) \;=\; +\infty.$$
- In this case, **Wald's lemma cannot be applied**

Example: Wald's Lemma Not Applicable

- Suppose $X_1, X_2, \cdots \overset{\text{i.i.d.}}{\sim} \mathcal{N}(1,1)$.
  For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.
  Let τ be defined as
  $$\tau := \inf\left\{n \geq 1 : S_n = \frac{\pi}{2}\right\}.$$
  Determine $\mathbb{E}[\tau]$.

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Kolmogorov's Existence Theorem and its Applications to Simulations

Observing a Random Phenomenon

- Suppose that you are observing a phenomenon through one or more measurements
- Suppose that **different measurements give rise to different values** for the quantity being measured
- It is natural to think of the measurements as different values of a **random variable** (say X)
- We do not typically have knowledge of the underlying $(\Omega, \mathscr{F}, \mathbb{P})$ giving rise to the measurements

Important Question

How do we know that there is an underlying $(\Omega, \mathscr{F}, \mathbb{P})$ and a measurable function $X : \Omega \to \mathbb{R}$ that fits our measurements?

- Often, we can estimate the **distribution (CDF)** of the measurements

Kolmogorov's Existence Theorem: Part 1

Theorem (Kolmogorov's Existence Theorem, Part 1)

Let $F : \mathbb{R} \to [0, 1]$ be any function satisfying the following properties:

1. $\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$
2. **Non-decreasing:** *If $x \leq y$, then $F(x) \leq F(y)$.*
3. **Right-continuity:** *For each $x \in \mathbb{R}$,*

$$\lim_{\varepsilon \downarrow 0} F(x + \varepsilon) \;=\; F(x).$$

Then, there exists a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and a function $X : \Omega \to \mathbb{R}$ such that:

1. *X is measurable with respect to $\mathscr{F}$.*
2. *For each $x \in \mathbb{R}$,*

$$\mathbb{P}(\{X \leq x\}) = F_X(x) = F(x).$$

Proof of Theorem 3

- The proof is simply a construction of an $(\Omega, \mathscr{F}, \mathbb{P})$ and X
- Let $(\Omega, \mathscr{F}, \mathbb{P}) = ((0,1),\ \mathscr{B}(0,1), \lambda)$, $\quad \lambda$: Lebesgue measure
- For any $\omega \in (0,1)$, let

$$X(\omega) \ = \ F^{-1}(\omega) \ := \ \inf\{x \in \mathbb{R} : \ F(x) \geq \omega\}.$$

  Clearly, X is a function with domain $\Omega = (0,1)$ and co-domain $\mathbb{R}$
- It is left as an exercise to show that X is measurable with respect to $\mathscr{F}$ and $F_X = F$
  Remark: This is the basic premise of the popular **inverse CDF technique**
  See, for instance, Lecture 21, AI5030 (Jul-Nov 2025), for more details

From a Random Variable to a Random Vector

```
array([ 0.52788774, -0.89551364,  0.78844006,  0.19032279,  0.92833631,
        0.00333179, -1.08593089, -0.52056583,  1.06936692, -0.51129284])
```

- We may want to map the above samples as

$$X_1(\omega) = 0.52788774, \quad X_2(\omega) = -0.89551364, \quad \ldots \quad X_{10}(\omega) = -0.51129284.$$

- **Use-case:** If the samples are drawn in an i.i.d. fashion from a distribution with a finite mean, then

$$\frac{1}{10}\sum_{n=1}^{10} X_n(\omega)$$

may be used as a proxy for the actual mean

Important Question

How do we know that there is an underlying $(\Omega, \mathscr{F}, \mathbb{P})$ and a measurable function $X = (X_1, \ldots, X_{10}) : \Omega \to \mathbb{R}^{10}$ that fits the samples?

Kolmogorov's Existence Theorem: Part 2

Theorem (Kolmogorov's Existence Theorem, Part 2)

Fix $n \in \mathbb{N}$. Let $F : \mathbb{R}^n \to [0, 1]$ be any valid multivariable CDF.
Then, there exists a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and $\mathbf{X} = (X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n$ such that:

1. *X is measurable with respect to $\mathscr{F}$.*
2. *For each $(x_1, \ldots, x_n) \in \mathbb{R}^n$,*

$$\mathbb{P}(\{X_1 \leq x_1, \ldots, X_n \leq x_n\}) \;=\; F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) \;=\; F(x_1, \ldots, x_n).$$

That is, $F_{X_1,\ldots,X_n} = F$.

Proof of Theorem 4

- Again, the proof is simply a construction of an $(\Omega, \mathscr{F}, \mathbb{P})$ and $\mathbf{X} : \Omega \to \mathbb{R}^n$
- Let $(\Omega, \mathscr{F}, \mathbb{P}) = (\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mu)$, where $\mu : \mathscr{B}(\mathbb{R}^n) \to [0, 1]$ is defined as:

$$\mu\Big((-\infty, x_1] \times \cdots \times (-\infty, x_n]\Big) \;=\; F(x_1, \ldots, x_n), \qquad (x_1, \ldots, x_n) \in \mathbb{R}^n,$$

  and $\mu(A)$ for any other $A \in \mathscr{B}(\mathbb{R}^n)$ is evaluated via Caratheodory's extension
- For any $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_n) \in \mathbb{R}^n$, let

$$X_1(\boldsymbol{\omega}) \;=\; \omega_1, \quad X_2(\boldsymbol{\omega}) \;=\; \omega_2, \quad \ldots \quad X_n(\boldsymbol{\omega}) \;=\; \omega_n.$$

  Let $\mathbf{X}(\boldsymbol{\omega}) = (X_1(\boldsymbol{\omega}), \ldots, X_n(\boldsymbol{\omega}))$
  Clearly, $\mathbf{X}$ is a function with domain $\Omega = \mathbb{R}^n$ and co-domain $\mathbb{R}^n$
- **Exercise:**
  Show that $\mathbf{X} = (X_1, \ldots, X_n)$ as defined above is measurable with respect to $\mathscr{F}$ and that $F_{X_1, \ldots, X_n} = F$

Handling a Sequence of Random Variables

- In many scenarios, we may not know apriori how many samples we need to generate

```
# Generate X1, X2, ... until their sum exceeds pi/2
import math
import random

S = 0.0
Xs = []
n = 0

while S <= math.pi / 2:
    n += 1
    X = random.gauss(0.0, 1.0)
    Xs.append(X)
    S += X
```

Handling a Sequence of Random Variables

- In many scenarios, we may not know apriori how many samples we need to generate
- We may want to map the samples as $X_1(\omega), X_2(\omega)$, etc., with terminal point not specified

Important Question

How do we know that there is an underlying $(\Omega, \mathscr{F}, \mathbb{P})$ and a sequence $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}} : \Omega \to \mathbb{R}^{\mathbb{N}}$ of measurable functions that fits the samples?

- Often, we have access to the **FDDs** of the sequence $\{X_n\}_{n\in\mathbb{N}}$ (e.g., i.i.d. process, Markov chain, martingale)

Kolmogorov's Existence Theorem: Part 3

Theorem (Kolmogorov's Existence Theorem, Part 3)

Let $\mathcal{T} = \mathbb{N}$.
Let $\mathbf{F} = \left\{ F_{\mathbf{t}} : \mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n, \;\; n \in \mathbb{N} \right\}$ *be a given collection of* **consistent** *FDDs.*

There exists a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ *and* $\mathbf{X} : \Omega \to \mathbb{R}^{\mathbb{N}}$ *such that:*

1. $\mathbf{X}$ *is measurable with respect to* $\mathscr{F}$.
2. *For each* $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n$, *and* $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$,

$$\mathbb{P}\big(\{X_{t_1} \leq x_1, \ldots, X_{t_n} \leq x_n\}\big) \;\; = \;\; F_{\mathbf{t}}(\mathbf{x}).$$

Proof of Theorem 5

- As before, the proof is simply a construction of an $(\Omega, \mathscr{F}, \mathbb{P})$ and $\mathbf{X} : \Omega \to \mathbb{R}^{\mathbb{N}}$
- Let $(\Omega, \mathscr{F}, \mathbb{P}) = (\mathbb{R}^{\mathbb{N}}, \mathscr{B}(\mathbb{R}^{\mathbb{N}}), \mu)$, where $\mu : \mathscr{B}(\mathbb{R}^{\mathbb{N}}) \to [0, 1]$ is any probability measure defined as:
  - For each $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n$, and $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$,
  $$\mu\Big(\Big\{\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots) \in \Omega : \quad \omega_{t_1} \leq x_1, \ldots, \omega_{t_n} \leq x_n\Big\}\Big) \quad = \quad F_{\mathbf{t}}(\mathbf{x}).$$
  - For each $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n$, and $B \in \mathscr{B}(\mathbb{R}^n)$, the value of
  $$\mu\Big(\Big\{\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots) \in \Omega : \ (\omega_{t_1}, \ldots, \omega_{t_n}) \in B\Big\}\Big)$$
  is evaluated via Caratheodory's extension for $\mathscr{B}(\mathbb{R}^n)$
  - **Kolmogorov's extension:** If FDDs in $\mathbf{F}$ are consistent, there exists a probability measure μ on $\mathscr{B}(\mathbb{R}^{\mathbb{N}})$ having the above coordinate-wise marginals
- For any $\boldsymbol{\omega} = (\omega_t)_{t \in \mathcal{T}} \in \mathbb{R}^{\mathbb{N}}$, let
$$X_t(\boldsymbol{\omega}) \quad = \quad \omega_t, \quad t \in \mathcal{T}, \qquad\qquad \mathbf{X}(\boldsymbol{\omega}) = (X_t(\boldsymbol{\omega}) : t \in \mathcal{T}).$$
- It can be argued that $\mathbf{X}$ is measurable with respect to $\mathscr{F}$ and that its FDDs match with $\mathbf{F}$

Kolmogorov's Existence Theorem: Part 4

Theorem (Kolmogorov's Existence Theorem, Part 4)

Let $\mathcal{T} = [0, T]$.

Let $\mathbf{F} = \left\{ F_{\mathbf{t}} : \mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n, \;\; n \in \mathbb{N} \right\}$ *be a given collection of* **consistent** *FDDs.*

There exists a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ *and* $\mathbf{X} : \Omega \to \mathbb{R}^{[0,T]}$ *such that:*

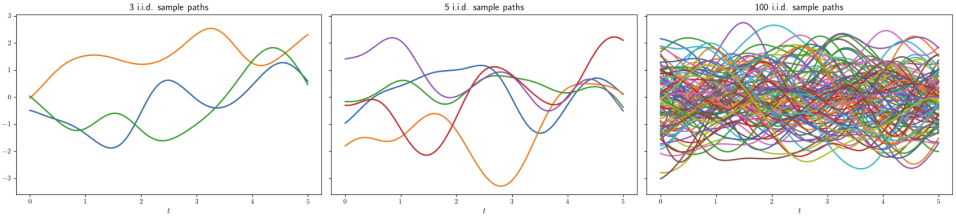
1. $\mathbf{X}$ *is measurable with respect to* $\mathscr{F}$.
2. *For each* $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in [0, T]^n$, *and* $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$,

$$\mathbb{P}\big(\{X_{t_1} \leq x_1, \ldots, X_{t_n} \leq x_n\}\big) \;\; = \;\; F_{\mathbf{t}}(\mathbf{x}).$$

Proof of Theorem 6

- As before, the proof is simply a construction of an $(\Omega, \mathscr{F}, \mathbb{P})$ and $\mathbf{X} : \Omega \to \mathbb{R}^{[0,T]}$
- Let $(\Omega, \mathscr{F}, \mathbb{P}) = (\mathbb{R}^{[0,T]}, \mathscr{B}(\mathbb{R}^{[0,T]}), \mu)$, where $\mu : \mathscr{B}(\mathbb{R}^{[0,T]}) \to [0, 1]$ is defined by:
  - For each $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n$, and $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$,
  $$\mu\Big(\Big\{\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots) \in \Omega : \quad \omega_{t_1} \le x_1, \ldots, \omega_{t_n} \le x_n\Big\}\Big) \quad = \quad F_{\mathbf{t}}(\mathbf{x}).$$
  - For each $n \in \mathbb{N}$, $\mathbf{t} = (t_1, \ldots, t_n) \in \mathcal{T}^n$, and $B \in \mathscr{B}(\mathbb{R}^n)$, the value of
  $$\mu\Big(\Big\{\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots) \in \Omega : \ (\omega_{t_1}, \ldots, \omega_{t_n}) \in B\Big\}\Big)$$
  is evaluated via Caratheodory's extension for $\mathscr{B}(\mathbb{R}^n)$
  - **Kolmogorov's extension:** If FDDs in $\mathbf{F}$ are consistent, there exists a probability measure μ on $\mathscr{B}(\mathbb{R}^{[0,T]})$ having the above coordinate-wise marginals
- For any $\boldsymbol{\omega} = (\omega_t)_{t \in [0,T]}$, let
$$X_t(\boldsymbol{\omega}) \quad = \quad \omega_t, \quad t \in [0.T], \qquad\qquad \mathbf{X}(\boldsymbol{\omega}) = (X_t(\boldsymbol{\omega}) : t \in [0, T]).$$
- It can be argued that $\mathbf{X}$ is measurable with respect to $\mathscr{F}$ and that its FDDs match with $\mathbf{F}$

Example in Action: Sampling from a GP



- Consider $\mathcal{T} = [0, 5]$. Fix $n \in \mathbb{N}$.
- Generate n i.i.d. samples from $\mathrm{GP}(M, C)$, where mean function M and autocovariance function C are

$$M : \mathcal{T} \to \mathbb{R}, \qquad M(t) = 0 \quad \forall\, t \in \mathcal{T},$$

$$C : \mathcal{T}^2 \to \mathbb{R}, \qquad C(s,t) = \underbrace{\sigma^2 \exp\left(-\frac{(t-s)^2}{2\ell^2}\right)}_{\text{RBF Kernel}}, \qquad \sigma = 1, \ \ell = 0.6$$

- **What model (probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and measurable function $\mathbf{X}$) fits the above simulation?**

Example in Action: Sampling from a GP

- $\Omega = \underbrace{\mathbb{R}^{[0,5]} \times \cdots \times \mathbb{R}^{[0,5]}}_{n \text{ times}}, \qquad \mathscr{F} = \mathscr{B}(\Omega) = \bigvee_{\ell=1}^{n} \mathscr{B}(\mathbb{R}^{[0,5]}), \qquad \mathbb{P} = \mu^{\otimes n}.$
- The probability measure $\mu : \mathscr{B}(\mathbb{R}^{[0,5]}) \to [0, 1]$ comes from Kolmogorov's extension (Theorem 6) of the FDDs of $\mathrm{GP}(M, \mathcal{C})$
- For any $A_1 \times \cdots \times A_n \in \mathscr{B}(\mathbb{R}^{[0,5]}) \times \cdots \times \mathscr{B}(\mathbb{R}^{[0,5]})$,

$$\mathbb{P}(A_1 \times \cdots \times A_n) = \prod_{\ell=1}^{n} \mu(A_\ell).$$

- For any $\boldsymbol{\omega} = (\omega_1, \ldots \omega_n) \in \Omega$, set

$$X^{(1)}(\boldsymbol{\omega}) = \omega_1, \qquad \ldots \qquad X^{(n)}(\boldsymbol{\omega}) = \omega_n, \qquad \mathbf{X}(\boldsymbol{\omega}) = \left(X^{(1)}(\boldsymbol{\omega}), \ldots, X^{(n)}(\boldsymbol{\omega})\right).$$

Stopping Times Continued

The Stopped Process

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ be a random process.
Let $\mathscr{G} = \{\mathscr{G}_n : n \in \mathbb{N}\}$ be the natural filtration adapted to the process $\mathbf{X}$.
Let τ be a stopping time adapted to $\mathscr{G}$. Define

$$\mathbf{Y}_\tau \;:=\; (X_1, \ldots, X_\tau), \qquad\qquad \mathbf{Y}_{\tau(\omega)}(\omega) \;=\; (X_1(\omega), \ldots, X_{\tau(\omega)}(\omega)), \quad \omega \in \Omega.$$

Then, $\mathbf{Y}_\tau$ is called the **stopped process**.

Important Question

Two important questions are of relevance.

- What is the codomain of $\mathbf{Y}_\tau$?
- If $\mathbb{R}^\star$ is the co-domain, what is $\mathscr{B}(\mathbb{R}^\star)$?

The Stopped Process

- The length of $\mathbf{Y}_\tau$ is not fixed
- Define $\mathbf{Z}_\tau$ as

$$\mathbf{Z}_\tau \;:=\; (\tau, \mathbf{Y}_\tau) \;=\; (\tau, X_1, \ldots, X_\tau).$$

  Notice that $\mathbf{Y}_\tau$ and $\mathbf{Z}_\tau$ have one-one correspondence
- Then, the codomain of $\mathbf{Z}_\tau$ is

$$\mathbb{R}^\star \;=\; \bigcup_{n\in\mathbb{N}} \Big(\{n\} \times \mathbb{R}^n\Big).$$

- Each element of $\mathbb{R}^*$ is of the form $(n, x_1, \ldots, x_n)$ for some $n \in \mathbb{N}$ and $x_1, \ldots, x_n \in \mathbb{R}^n$
- The Borel σ-algebra on $\mathbb{R}^\star$ is given by

$$\mathscr{B}(\mathbb{R}^\star) \;=\; \Big\{A \subseteq \mathbb{R}^\star : \quad \underbrace{A \cap \Big(\{n\} \times \mathbb{R}^n\Big)}_{\text{slice of dimension } n} \in \mathscr{B}(\{n\} \times \mathbb{R}^n) \quad \forall\, n \in \mathbb{N}\Big\}.$$

- The generating class for $\mathscr{B}(\mathbb{R}^\star)$ is given by

$$\mathscr{E} \;=\; \Big\{\{n\} \times B : \quad n \in \mathbb{N},\;\; B \in \mathscr{B}(\mathbb{R}^n)\Big\}.$$

The Stopped Process

- We now prove that $\mathbf{Z}_\tau : \Omega \to \mathbb{R}^\star$ is indeed a random variable
- To prove that $\mathbf{Z}_\tau$ is a random variable, it suffices to show that for any $B^\star \in \mathscr{E}^\star$,

$$\mathbf{Z}_\tau^{-1}(B^\star) \in \mathscr{F}.$$

- Taking $B^\star = \{n\} \times B$ for fixed $n \in \mathbb{N}$ and $B \in \mathscr{B}(\mathbb{R}^n)$, we have

$$\mathbf{Z}_\tau^{-1}(\{n\} \times B) \;=\; \{\tau = n\} \cap \{(X_1, \ldots, X_n) \in B\} \;\in\; \mathscr{G}_n \;\subseteq\; \mathscr{F}.$$

What is $\sigma(\mathbf{Z}_\tau)$?

The Stopped σ-Algebra

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Definition (Stopped σ-Algebra)

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a random process.
Let $\mathscr{G} = \{\mathscr{G}_n : n \in \mathbb{N}\}$ be the natural filtration adapted to the process $\mathbf{X}$.
Let τ be a stopping time adapted to $\mathscr{G}$. Then,

$$\mathscr{G}_\tau \;:=\; \Big\{ E \in \mathscr{F} : \quad E \cap \{\tau = t\} \in \mathscr{G}_t \quad \forall\ t \Big\}$$

is called the **stopped σ-algebra**.

Remark:

- $\mathscr{G}_\tau$ is a σ-algebra of subsets of Ω
- If $\mathcal{T} = \mathbb{R}$ or $\mathbb{R}_+$, then

$$\{\tau = t\} \;\leftarrow\; \{\tau \leq t\}$$

  in the definition of the stopped σ-algebra

Strong Independence Property

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Lemma (Strong Independence Property)

Let $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ be an **independent process** *with respect to $\mathscr{F}$.*
Let τ be a **stopping time** *adapted to the natural filtration of the process $\mathbf{X}$. Then, for any $n \in \mathbb{N}$,*

$$\mathbb{P}\Big(\{X_{\tau+n} \leq x\} \mid \tau, X_1, \ldots, X_\tau\Big) \;=\; \mathbb{P}\Big(\{X_{\tau+n} \leq x\} \mid \tau\Big).$$

That is, conditioned on the stopping time, future is independent of the past and present.

Remark:

- For any (deterministic) $n \in \mathbb{N}$, we know

$$(X_1, \ldots, X_n) \;\perp\!\!\!\perp\; (X_{n+1}, X_{n+2}, \ldots).$$

- Strong independence property says that the above is true if n is replaced by a stopping time τ